

INTERNSHIP REPORT

THEME:

SIMULATION OF THE SEA SURFACE AS SEEN BY SATELLITE RADARS

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ABSTRACT

This internship, conducted from September 1st, 2025, to January 16th, 2026, at ICAM Nantes, focused on the development of a realistic monostatic radar simulator for the sea surface. The main objective was to integrate and compare 22 NRCS (Normalized Radar Cross Section) models – semi-empirical, empirical and physical – within a single graphical interface, while adding realistic 3D sea surface generation based on the Elfouhaily spectrum.

The simulator was implemented in MATLAB using uifigure and the GUI Layout Toolbox, offering two main modules: one dedicated to NRCS calculation and multi-curve visualization (with interactive overlay and legend), and the other to 3D sea surface synthesis with memory optimization and multi-format export (csv, .mat, .png).

Key achievements include automatic parameter validation, dynamic model filtering by frequency band, advanced memory management (RAM detection and 10% limit), resolution of graphical stability issues, and scientific validation of several models against the MATLAB Radar Toolbox and reference publications.

The tool provides a unified, educational and extensible platform for radar-sea interaction studies, supporting research, teaching and future extensions (bistatic configurations, parallel computing).

This work capitalized on more than twenty years of laboratory research, delivering a robust and user-friendly simulation environment for ocean remote sensing applications.

ACKNOWLEDGMENTS

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INTRODUCTION

As part of my 4^e year of computer engineering training, I completed a scientific research internship at ICAM, a site in Nantes, France. This internship, which took place from September 1^{er}, 2025 to January 16, 2026, allowed me to conduct an applied research experiment aimed at deepening my scientific and technical knowledge in a field at the crossroads of computer science, physics and digital modelling.

The theme of this course was "Simulation of the surface of the sea seen by satellite radars." This topic is part of the broader framework of radar remote sensing, a key area for Earth observation, maritime monitoring and understanding of the interactions between electromagnetic waves and the ocean surface. The main objective was to develop a realistic digital simulator of the radar response of a sea surface, based on physical parameters such as sea state, wind speed and direction, angle of incidence, polarization and radar frequency, relying on the MATLAB Radar Toolbox.

During this course, I carried out a bibliographic review of existing models (empirical, semi-empirical and physical), designed and implemented a complete graphic interface in two modules (calculation and visualization of the NRCS/generation and 3D display of sea surfaces), integrated 22 models NRCS, developed advanced features (interactive overlay, memory management, exports).

This immersion also allowed me to integrate fully into the ICAM research environment, to exchange regularly with my supervisor on methodology and scientific guidelines, and to contribute concretely to the capitalization of the laboratory's work.

The purpose of this internship report is to present all the work carried out, analyse the results obtained, draw up a technical and personal report and propose prospects for improvement. It is organized around the presentation of the hosting structure, the context and objectives of the project, the detailed technical realization, the validations and tests carried out, and then a general conclusion.

CHAPTER 1: GENERAL PRESENTATION OF THE COMPANY

1. History

The Catholic Institute of Arts and Crafts (ICAM) was founded in 1898 in Lille, on the initiative of industrialists and Jesuits of the North region, to meet the growing need for engineers in a context of rapid industrialization. Since then, ICAM has grown strongly, increasing the number of campuses in France and internationally, and establishing itself as a recognized general engineering school.

Concerning the West site, the ICAM campus in Nantes was created in 1988. It is part of the "ICAM Ouest" cluster, which also includes the Vendée (La Roche-sur-Yon) and Brittany (Vannes) sites, which were legally reunited in 2018 to strengthen their regional roots and synergies.

The creation of the Nantes campus served a twofold purpose: to extend the geographical presence of ICAM in the West of France and to meet the industrial and technological needs specific to this region, particularly in the aeronautics, naval and mechanical sectors.

Today, the Nantes campus welcomes several hundred engineering students, develops applied research activities and builds numerous regional partnerships to train engineers capable of meeting the challenges of industry and the technological transition.

2. Company Profile

Table 1: Company Profile

Name of company	ICAM Nantes site
Year of creation	1988
Head office	35 Avenue du Champ de Manoeuvre, 44470 Carquefou, France

Website	www. icam.fr
Electronic mail	icam-nantes@icam.fr
Telephone	+33 2 40 52 40 52
Director General	Jérôme LABARRE
Legal form	Association Declared
Enrolment	About 600 engineering students and 150 employees

3. Flowchart

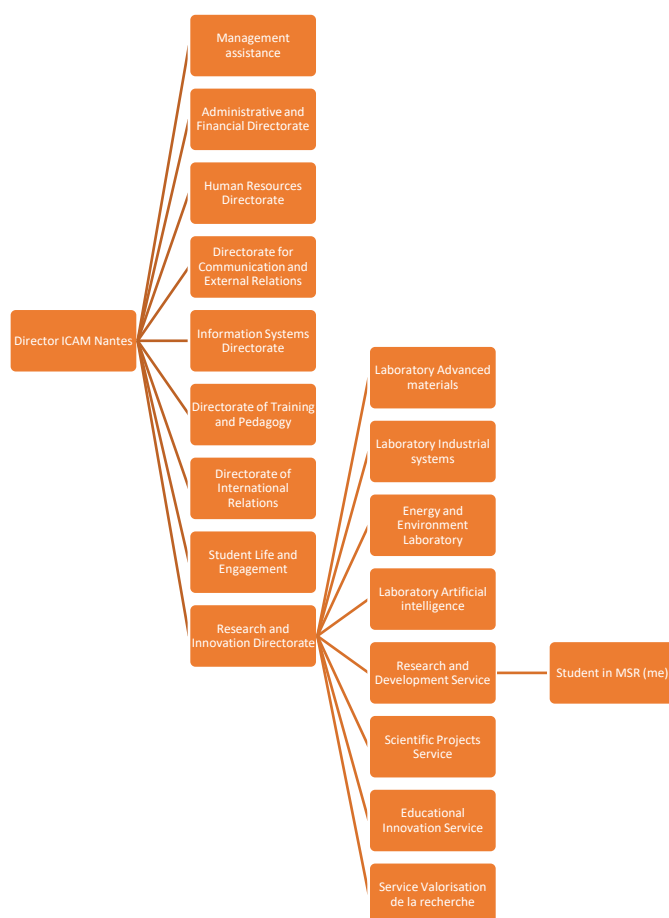


Figure 1: Organization chart of the enterprise

4. Location

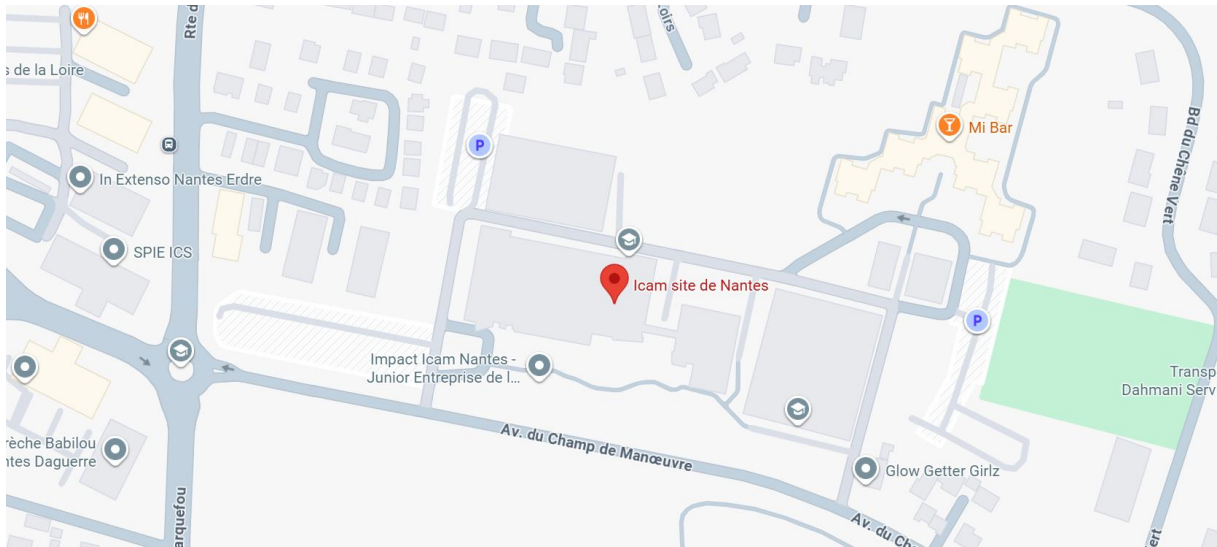


Figure 2: Location of the company

5. Corporate culture

a. Values

ICAM is distinguished by a deeply human and open culture, inherited from its Jesuit foundation. Its values are based on the integral training of the person, solidarity, responsibility, and a sense of service.

The school aims to train engineers who are both technically competent and humanely committed, able to reconcile performance and ethics in their professional decisions.

The spirit of collaboration, listening, and the search for meaning in work are an integral part of his identity. These values are reflected in educational projects, community life, but also in research and innovation carried out by teachers and students.

b. Working atmosphere

The working atmosphere within ICAM Nantes is friendly, collaborative and conducive to learning. Relationships between students, teachers and technical staff are based on trust and caring.

As part of the scientific research mission, the activities take place mainly in laboratories, where students from different classes live together. I have the opportunity to work alongside 5^e year

students in Industrial Scientific Mission (MSI), which encourages exchanges, the sharing of experiences and collaboration on various technical issues.

This proximity between the different levels of study and the tutors creates a stimulating environment, where autonomy and scientific rigour are encouraged while benefiting from attentive support. The general atmosphere combines seriousness, mutual assistance and intellectual curiosity, ideal conditions for the success of an applied research project.

CHAPTER 2: PRESENTATION OF THE PROJECT

1. Background to the project

Maritime surveillance is a major strategic issue for many applications, such as border security, environmental protection and the monitoring of naval activities. Radars play a central role in this surveillance, detecting and tracking ships, waves and other objects on the surface of the sea.

Interpretation of radar data, however, remains complex because it depends on multiple factors, such as weather conditions, the state of the sea or the specific characteristics of the radar used. In this context, the development of a simulator of the surface of the sea seen by a radar is of considerable scientific and operational interest. Such a simulator makes it possible to better understand the interactions between radar waves and the ocean surface, to evaluate the performance of existing radar systems, and to design signal processing algorithms that are more efficient. It is also an educational tool for training radar operators and testing new configurations.

2. Challenges and stakes

Scientific and operational stakes

- Provide the laboratory with a comprehensive, flexible simulation platform capable of capitalizing on two decades of model development,
- Facilitate the comparison and validation of semi-empirical, empirical, and physical models across a wide range of radar configurations,
- Support future research on advanced configurations (highly agitated seas, bistatic radar...),
- Serve as a reference tool for operational radar performance evaluation and algorithm design.

Personal and academic stakes

- Acquire advanced skills in graphical interface development (MATLAB uifigure + GUI Layout Toolbox),
- Understanding of electromagnetic scattering models and sea surface physics,

- Develop rigorous software engineering practices (modularity, robustness, memory optimization),
- Contribute to the laboratory's research continuity while producing a concrete, shareable deliverable.

3. Objectives of the project

The project aims to **develop a realistic simulator of the surface of the sea seen by a radar**, considering the different factors influencing radar data. This simulator is based on:

- Physical models describing the reflection of radar waves by the surface of the sea,
- Experimental models based on real measurements.

The specific objectives are:

- **Develop a graphical parameterization interface** using the functionality of the MATLAB Radar Toolbox, allowing to select the type of radar (PPI, Doppler...), to define the simulation parameters and to visualize the results.
- **Incorporate the electromagnetic models developed within the laboratory**, enriching the simulator with advanced features, such as considering highly agitated surfaces or bistatic configurations.
- **Visualize and compare simulated data with actual data** to assess the accuracy and reliability of the models used.
- Depending on the progress, **implement a simplified two-dimensional version** for fast and exploratory simulations.

This project is a continuation of the research carried out for more than twenty years in the laboratory, aimed at modelling the reflection of electromagnetic waves by the surface of the sea and capitalizing on all the codes and methods already developed. The completion of this project will provide a comprehensive, flexible simulation tool adapted to scientific studies as well as practical applications in the radar field.

CHAPTER 3: STUDY AND SOLUTION

1. State of the art of models NRCS

The study of sea radar reflectivity is based on a set of empirical, semi-empirical and physical models developed over several decades. Within the laboratory, several families of models are implemented to cover different radar frequencies, observation geometries and sea conditions. These models make it possible to represent the dependence of **Normalized Radar Cross Section** on geophysical parameters: wind speed, wind direction, sea state, polarization or incidence.

Semi-Empirical models

These models combine theoretical considerations (e.g., scattering mechanisms at low grazing angles) with empirical adjustments derived from experimental data. They are particularly effective for high incidence angles (grazing regime) and are widely used in military and coastal surveillance applications.

- NRL (0.5–35 GHz, 30–89.9°, HH/VV, u_{10} 0–13.2 m/s)
- TSC (0.5–35 GHz, 0–89.9°, HH/VV, u_{10} 0–11.4 m/s)
- GIT (1–100 GHz, 80–89.9°, HH/VV, u_{10} 3.2–13.2 m/s)
- HYBRID (0.1–100 GHz, 60–90°, HH/VV, u_{10} 0–11.4 m/s)
- SITPROP (9.375 GHz, 80–89.8°, HH/VV, u_{10} 0–14.9 m/s, wind angle 0° or 90)
- RRE (9–10 GHz, 80–89.9°, HH/VV, u_{10} 3.2–13.2 m/s)

Experimental models

These are purely data-driven models, adjusted on large datasets from airborne or satellite measurements. They are the reference for operational applications (scatterometers, altimeters) but remain limited to specific radar configurations.

L band (1–2 GHz)

- Isoguchi (1.3 GHz, 17–43°, HH/VV, u_{10} 0–20 m/s, wind angle 0–360°)
- Meissner (1.26 GHz, 24.4–49.3°, VV/HH/HV, u_{10} 0.2–22 m/s, wind angle 0–360°)

C band (4–8 GHz)

- Quilfen (5.3 GHz, 18–60°, VV only, $u_{10} < 20$ m/s, wind angle 0–360°)
- CMOD2-I3 (5.3 GHz, 18–57°, VV only, u_{10} 3–25 m/s, wind angle 0–360°)
- CMOD5 (5.3 GHz, 18–57°, VV only, u_{10} 0–25 m/s, wind angle 0–360°)

X band (8–12 GHz)

- XMOD1R (10 GHz, 20–60°, VV only, u_{10} 0–20 m/s, wind angle 0–360°)
- XMOD2N (10 GHz, 18–46°, VV only, u_{10} 2–25 m/s, wind angle 0–360°)
- Masuko_X (10 GHz, 30–60°, HH/VV, u_{19} 3.2–17.2 m/s, wind angle 0–360°)

Ku band (12–18 GHz)

- Wentz84 (14.6 GHz, 0–60°, HH/VV, u_{10} 0–20 m/s, wind angle 0–360°)
- Wentz99 (14.6 GHz, 15–65°, HH/VV, u_{10} 0–20 m/s, wind angle 0–360°)
- NSCAT1 (13.6 GHz, 15–60°, HH/VV, u_{10} 1–30 m/s, wind angle 0–360°)

Ka band (27–40 GHz)

- Masuko_Ka (34.43 GHz, 30–60°, HH/VV, u_{19} 3.2–17.2 m/s, wind angle 0–360°)

Statistical models

These models are derived from electromagnetic scattering theory and are more general (broad frequency range, multiple polarizations). They explicitly depend on the sea surface spectrum (e.g., Elfouhaily) and environmental parameters (fetch, permittivity).

- TSM (0.1–100 GHz, 0–80°, VV/HH/HV, u_{10} 1–25 m/s, wind angle 0–360°, fetch 10^3 m – Inf)
- SSA1 (0.1–100 GHz, 0–80°, VV/HH/HV, u_{10} 1–25 m/s, wind angle 0–360°, fetch 10^3 m – Inf)

2. Critic of the existing situation

While these models are scientifically sound within their validity domains, their practical use remains limited by:

- Dispersion of implementations (isolated scripts or functions)
- Lack of a unified graphical interface for comparison and visualization

- No interactive superposition of multiple models
- Limited robustness (no automatic parameter validation, no memory management)

3. Proposed solution and improvements

To overcome these limitations, the project developed a unified and interactive simulator that integrates all 22 models into a single, user-friendly tool.

Main improvements brought:

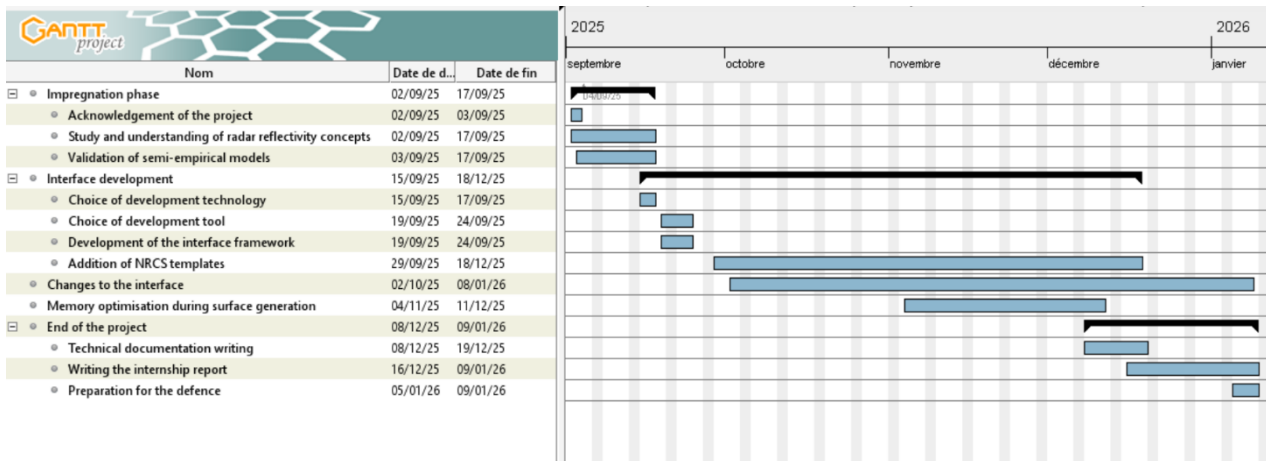
- **Graphical interface** structured in two modules: NRCS calculation/visualization and realistic sea surface generation
- **Dynamic parameter management** with automatic validation of validity ranges
- **Interactive multi-curve superposition** with clickable legend for deletion
- **3D sea surface generation** based on the Elfouhaily spectrum with fetch dependency
- **Advanced features:** memory optimization (RAM detection, 10% limit), multi-format exports (.csv, .mat, .png), progress dialog
- **Modularity and extensibility** for future additions (bistatic, new models...)

This solution transforms scattered models into a coherent, educational and operational tool, facilitating comparison, exploration and validation of radar-sea interaction models.

CHAPTER 4: PROJECT MANAGEMENT

1. Planning

a. Actual schedule



2. Budget

Designation	Characteristics / Description	Quantity	Unit Price	Total Price	References / Justification
Material Resources					
Lenovo ThinkBook 14s YogaG3 IRU	Laptop used for development and testing	1	0 € (already owned)	0 €	ThinkBook 14s Yoga Gen 3 Puissant portable 2-en-1 pour PME de 35,56 cm (14") équipé d'un processeur Intel Lenovo France
Software Resources					
MATLAB (student/academic license)	Main development environment + Radar Toolbox + GUI Layout Toolbox	1	85 € (student rate)	85 €	Prix et licences - MATLAB & Simulink
Windows 11 Pro	Operating system of the computer	1	0 € (pre-installed)	0 €	Achetez et téléchargez Windows 11 Pro Microsoft Store

Human Resources					
Junior software developer	Full-time work on design, implementation, testing and reporting (4.5 months $\approx$ 18 weeks $\approx$ 720 hours)	1	18,54 € / hour	13 348,80 €	Salaire développeur junior - Vérifier le salaire moyen développeur junior sur Jooble
TOTAL				13 398,80 €	

CHAPTER 5: IMPLEMENTATION OF THE SOLUTION

1. Choice of technologies

a. Development environment

Table 2: Comparative table of technological solutions

Criterion	MATLAB	Python + MATLAB Engine	C/C++ + MATLAB Engine
Ease of development	Very simple, fully integrated	Requires MATLAB Engine setup and two languages	Complex, memory management and Engine API
Graphical interface (UI)	Basic (App Designer)	Modern and flexible (PyQt, Tkinter, Streamlit, Web)	Powerful (Qt/C++) but heavy to develop
Performance	Good for prototyping	Overhead due to Engine calls	Very good, close to hardware
Interoperability	Limited to MATLAB and its toolboxes	Broad (ML, Web, Data Science in Python)	Good for hardware and real-time applications
Maintenance	Simple (single environment)	Medium (shared MATLAB + Python code)	Difficult, high cost
MATLAB License	Required	Required	Required
Ideal use case	Prototyping, research	Modern interface combined with MATLAB Radar Toolbox	Industrial applications, embedded, defense

This comparison highlights MATLAB's clear advantage for rapid prototyping and research purposes, justifying its selection as the primary development environment for the simulator.

b. Graphical interface development tool

Table 3: Comparison of Graphical Interface Development Approaches

Criterion	GUI Layout Toolbox	App Designer
Ease of use	Manual coding	Drag-and-drop
Flexibility	Highly customizable	Limited, less modular

Performance	Good for large-scale projects	Better for visualization
Deployment	Easy	Native to MATLAB
Compatibility	R2014a+	R2016a+

The GUI Layout Toolbox was chosen over App Designer for its superior flexibility and compatibility with older MATLAB versions, allowing a highly tailored and robust user interface.

2. Simulator architecture

The simulator is implemented as a single executable MATLAB script (SeaRadarSimulator.m) for ease of deployment and maintenance.

Overall structure

- Creation of the main window using uifigure
- Central data structure app handles storing all UI components and runtime data (NRCS curves, current surface)
- Two tabs managed by uitabgroup:
 - **Tab 1 – NRCS Simulation** → calculation and visualization of backscattering
 - **Tab 2 – Sea Surface Generation** → synthesis and 3D display of realistic sea surfaces

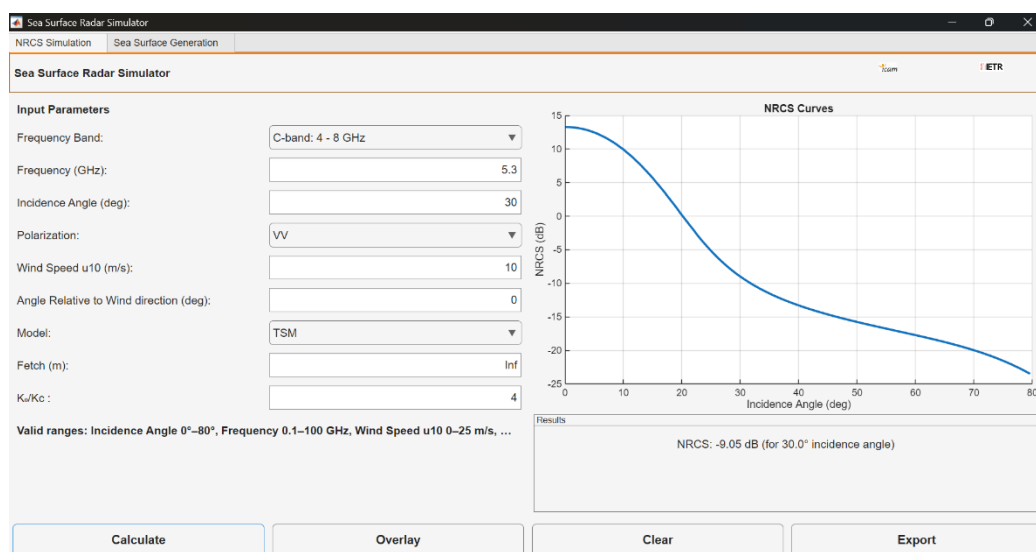


Figure 3: Tab 1 – NRCS Simulation

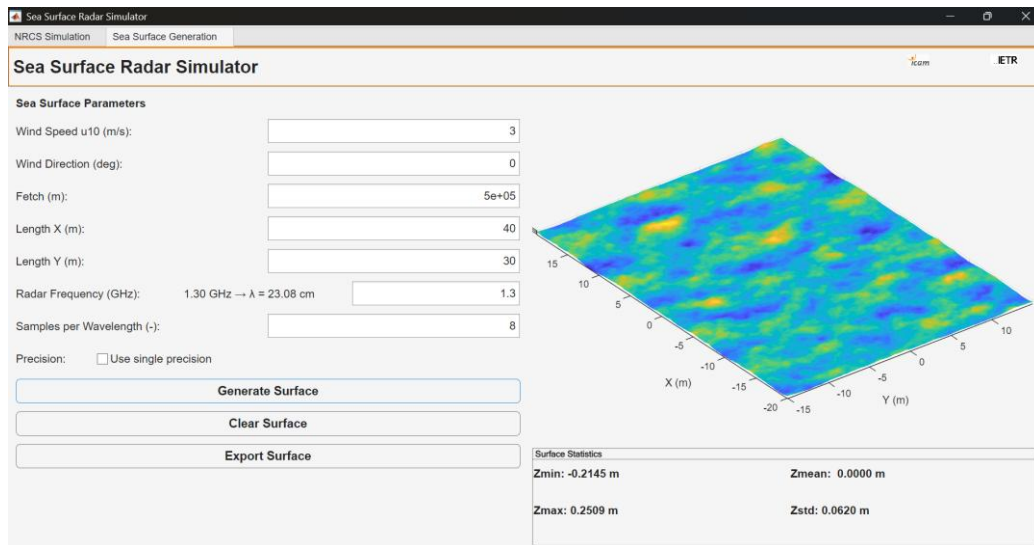


Figure 4: Tab 2 – Sea Surface Generation

The architecture relies on modular callbacks (independent functions triggered by user actions) and a centralized data structure, ensuring code readability, maintainability, and ease of future extensions.

3. Graphical interface design

The graphical interface was designed to be intuitive, ergonomic, and responsive, with a clear separation between input parameters and result visualization.

Common header (both tabs)

- Orange banner with simulator title (22 pt font)
- Dynamic loading and resizing of ICAM and IETR logos (fallback text if images are missing)

Tab 1 – NRCS Simulation

- **Left panel (parameters):**
 - Frequency band selection (UHF to W-band) with automatic central frequency update
 - Radar frequency (GHz)
 - Incidence angle (°)
 - Polarization (HH, VV, HV)
 - Wind speed u_{10} (0–25 m/s)

- Relative wind angle (0–360°)
- Fetch (enabled only for TSM, SSA1, GO1sh, SPM1)
- K_0/K_c ratio (enabled for TSM and GO1sh)
- Automatic wind-to-sea state conversion for semi-empirical models
- **Right panel (display and results):**
 - Main uiaxes for NRCS curve plotting
 - Results panel showing NRCS value at selected incidence angle
 - Warning label (orange) for approximations or limits
- **Bottom buttons:** Calculate, Overlay, Clear, Export

Tab 2 – Sea Surface Generation

- **Left panel (parameters):**
 - Wind speed u_{10} (0–25 m/s)
 - Wind direction (0–360°)
 - Fetch
 - Surface dimensions (X and Y in meters)
 - Radar frequency (GHz) with real-time wavelength display
 - Samples per wavelength
 - Precision checkbox (single/double)
- **Right panel:** 3D visualization of the generated surface
- **Bottom buttons:** Generate Surface, Clear Surface, Export Surface

4. Integration of NRCS models

A total of **22 NRCS models** were integrated, covering the three main families:

- **Semi-empirical** → NRL, TSC, GIT, HYBRID, SITPROP, RRE
- **Empirical** → Quilfen, CMOD2-I3, CMOD5N, Isoguchi, XMOD1R, XMOD2N, Meissner, Wentz84, Wentz99, Masuko_X, Masuko_Ka, NSCAT1
- **Physical/statistical** → TSM, SSA1, GO1sh, SPM1

Each model is selected via a dynamic dropdown (filtered by frequency band). The calculation function (*calculusCallback*) includes:

- Automatic parameter validation (incidence angle, frequency, wind speed, polarization)
- Specific vector θ generation per model
- Model-specific function calls (switch-case structure)
- Conversion to dB for linear-scale models
- Storage of up to 4 curves in *app.handles.curves*

5. Advanced features

- **Interactive overlay** (*superposerCallback*)
 - Up to 4 curves displayed simultaneously with distinct styles/colours
 - Clickable legend for curve deletion
 - Dynamic Y-axis limits (especially adapted for GO1sh)
 - Enriched DataTips
- **Sea surface generation** (*generateSeaSurfaceCallback*)
 - 3D synthesis based on *Elfouhaily* spectrum with fetch dependency
 - RAM detection and blocking
 - Single/double precision option
- **Exports** (*exporterCallback*, *exportSeaSurfaceCallback*)
 - NRCS curves: single or multi-column CSV
 - Sea surfaces: .mat (full structure), .csv (X;Y;Z), .png (600 dpi), or all three at once
- **Memory optimization**
 - Automatic detection of available RAM (multi-platform: Windows, Linux, MacOS)
 - Limit set to 10% of total memory, blocking with detailed alert if exceeded

INTERNSHIP REPORT

1. Review

Although still ongoing, the main objectives of the project have been largely achieved. The simulator developed integrates twenty NRCS models (semi-empirical, empirical and statistical) within a graphical interface structured into two distinct modules: the first dedicated to calculating and visualizing the NRCS according to the models implemented, the second enabling the generation and display of realistic 3D sea surfaces based on the Elfouhaily spectrum. The interface offers features such as multi-curve overlay, data export, automatic parameter validation and optimized memory management.

2. Personal opinion

This internship was very rewarding. It allowed me to deepen my skills in graphical interface development, discover MATLAB and radar reflectivity modelling. Overcoming challenges related to graphical stability and memory management strengthened my problem-solving skills and perseverance. Although this project was a little complicated at first because of the complexity of the subject matter, it allowed me to test my adaptability once again and proved to be a very stimulating experience.

3. Outlook

- Extend the availability of SSA1 correlation files for non-infinite fetch cases.
- Exploring task parallelisation using the Parallel Computing Toolbox to add a third tab to the simulator
- Managing interface responsiveness issues
- Switch to a bistatic configuration with different input parameters for the transmitter and receiver

CONCLUSION

This scientific research internship at ICAM Nantes is a decisive step in my academic and professional career. Working on the development of a realistic simulator of the surface of the sea viewed by radar allowed me to plunge into the heart of a project with a strong scientific and technological dimension, while closely linking the theoretical knowledge acquired in training to their concrete application.

Over the months, I have considerably deepened my skills in developing modern graphical interfaces under MATLAB (uifigure, GUI Layout Toolbox), in modeling radar backscattering (integration of 22 models NRCS), in generating realistic ocean surfaces (Elfouhaily spectrum) and in software optimization (memory management, robustness). I also discovered the methodological rigour and experimental approach specific to applied research.

Mr. Nicolas PINEL 'careful supervision, regular exchanges with 5^e year students and researchers, as well as the collaborative atmosphere of the team allowed me to gain a great deal of autonomy, to structure my scientific reasoning and to overcome concrete technical challenges (graphic stability, management of the RAM, validation of models).

This internship, fully completed, offered me much more than a simple professional experience: it confirmed my vocation for professions at the interface between computer science, applied physics and oceanographic research. He also endowed me with solid and transferable skills, both technical and human, which will be precious to me in the rest of my career.

I would like to thank once again all those who have made this mission possible and enriching. This report marks the end of a great adventure, but also the beginning of a lasting commitment in the fields of remote sensing and scientific simulation.

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APPENDIX

APPENDIX 1: Glossary

This glossary lists and explains the main technical terms used in the report, particularly those related to MATLAB programming, graphical user interface development, radar physics and sea surface modeling.

- **app.handles** Global MATLAB structure used to store all user interface components (buttons, fields, axes, etc.) as well as computed data (NRCS curves, current sea surface). It allows easy access to interface elements from any callback function.
- **Callback** MATLAB function automatically called when a user event occurs (button click, value change, window resize). Examples: `calculusCallback`, `superposerCallback`, `bandChangedCallback`.
- **Dropdown (uidropdown)** Graphical user interface component (drop-down menu) allowing selection of an option from a predefined list. Used for frequency band, NRCS model or polarization selection.
- **Fallback** Backup or default mechanism. In this code, it refers to the text "ICAM | IETR" displayed if the logo image files (`logo_icam.png`, `logo_ietr.png`) are missing or cannot be loaded.
- **GUI Layout Toolbox (uix)** External MATLAB toolbox (free Add-On) that enables flexible and responsive layout design without using App Designer. It provides containers such as `uix.VBox`, `uix.HBox`, `uix.Grid` for organizing components.
- **uiaxes** Modern MATLAB graphics component (equivalent to classic axes but designed for `uifigure`). Used to display NRCS curves and 3D sea surfaces.
- **uibutton** Graphical button component of the user interface. Examples: "Calculate", "Overlay", "Generate Surface".
- **uicheckbox** Checkbox allowing activation/deactivation of an option (e.g., single/double precision for surface generation).
- **uieditfield (numeric)** Numeric input field allowing the user to enter values (frequency, incidence angle, wind speed, etc.).
- **uilabel** Static text component displayed in the interface (titles, validity notes, wavelength display, etc.).
- **uifigure** Modern MATLAB main window (equivalent to classic figure but for App Designer/`uifigure`-based interfaces). It is the root window of the simulator.
- **uitab / uitabgroup** Tabs and tab group components. The simulator uses one `uitabgroup` containing two `uitab`: "NRCS Simulation" and "Sea Surface Generation".

- **uix.HBox / uix.VBox** Horizontal (HBox) and vertical (VBox) containers from GUI Layout Toolbox. They automatically align components side-by-side (horizontally) or stacked (vertically) with controlled spacing and padding.
- **uix.Grid** Grid (table-like) container from GUI Layout Toolbox, used here to place the graphics axes (uiaxes) and occupy the full available area.
- **uialert** Modern MATLAB pop-up alert or information window (more advanced than msgbox). Used to display warnings and success messages.
- **uiprogressdlg** Progress dialog box (indeterminate loading bar) used during sea surface generation.
- **NRCS (Normalized Radar Cross Section) / σ^0** Normalized radar cross-section (in dB), the key physical quantity measuring the power backscattered per unit area of the sea surface.
- **Fetch** Distance over which the wind blows consistently (in meters or km). It influences wave development and therefore surface roughness.
- **u₁₀** Wind speed at 10 meters above sea level (in m/s), main input parameter for empirical and semi-empirical models.
- **Incidence angle θ** Angle between the radar beam and the normal to the sea surface
- **Relative wind angle** Angle between wind direction and radar beam direction (0° – 360°).
- **Polarization** Orientation of the radar electromagnetic wave electric field: VV (vertical transmit/vertical receive), HH (horizontal), HV (cross-polarized).
- **Elfouhaily spectrum** Unified directional spectrum of wind-generated waves (long and short waves), published in 1997, used as the basis for realistic sea surface generation in the simulator.

APPENDIX 2: Interface images

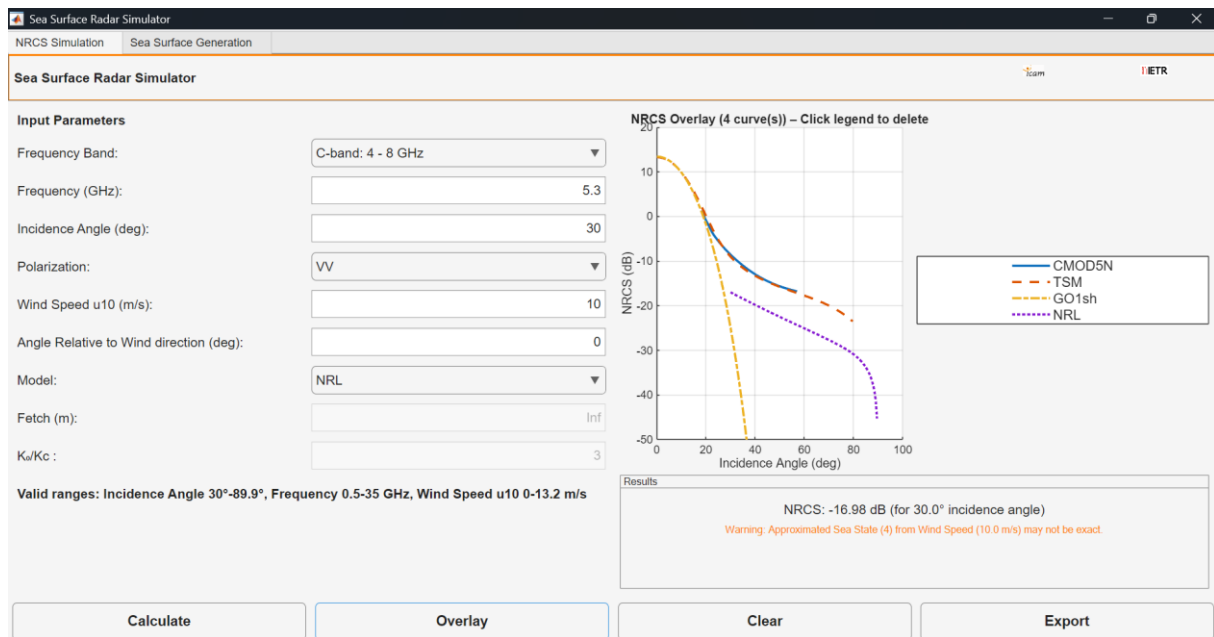


Figure 5: Tab 1 - Overlay

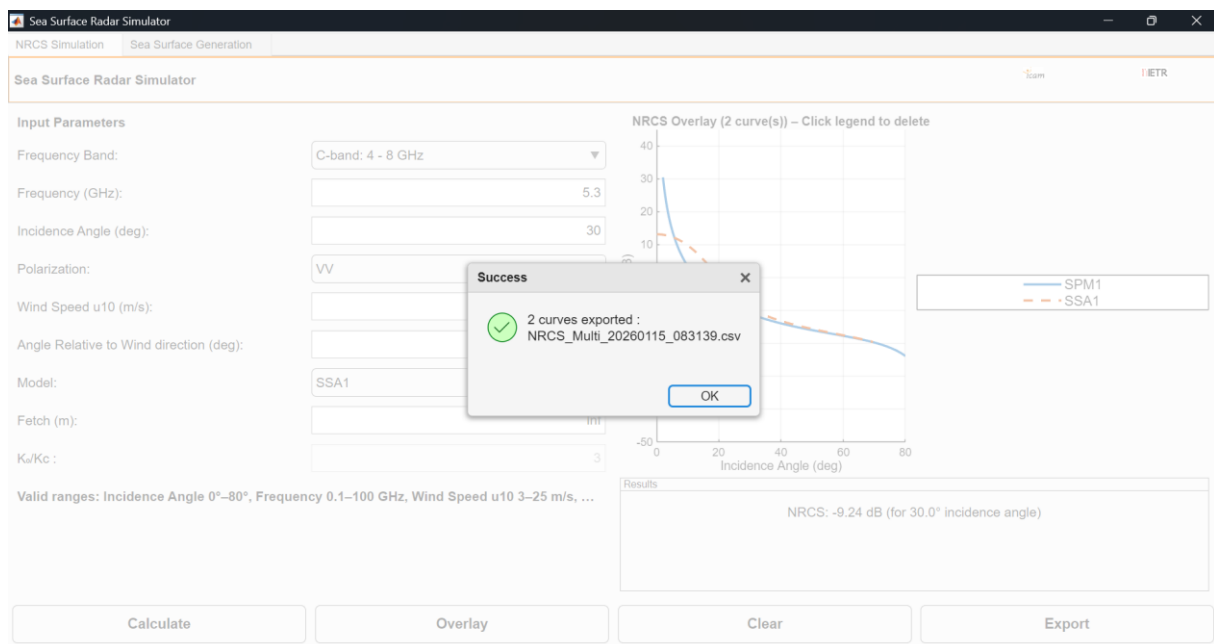


Figure 6: Tab 1 - Export

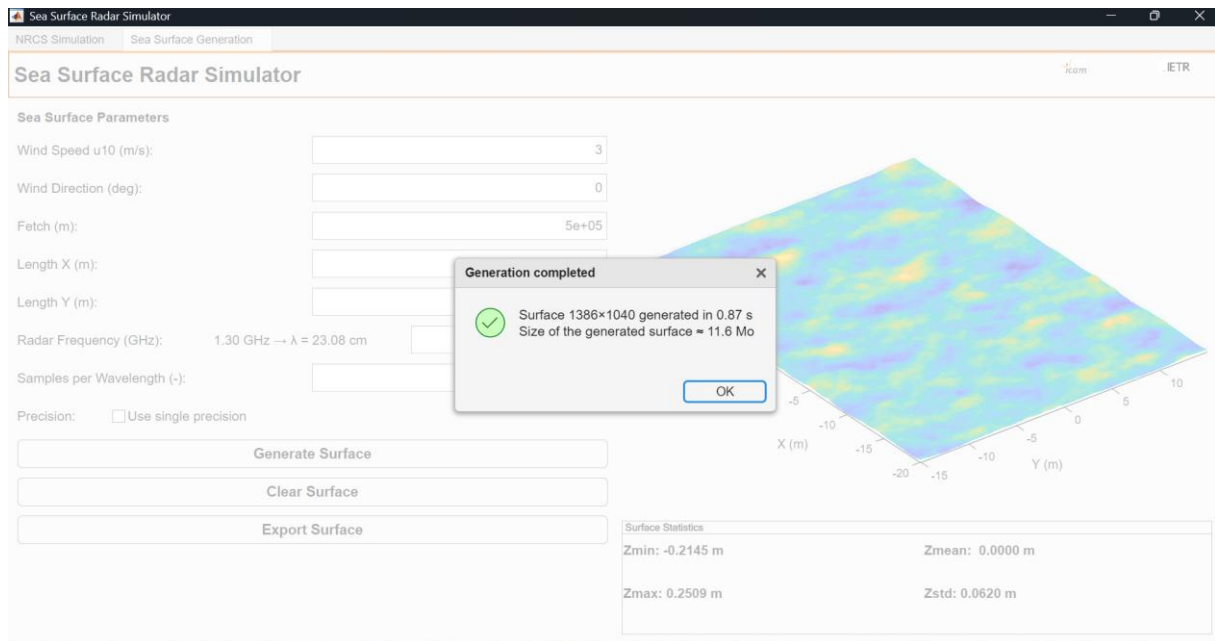


Figure 7: Tab 2 - Surface generation

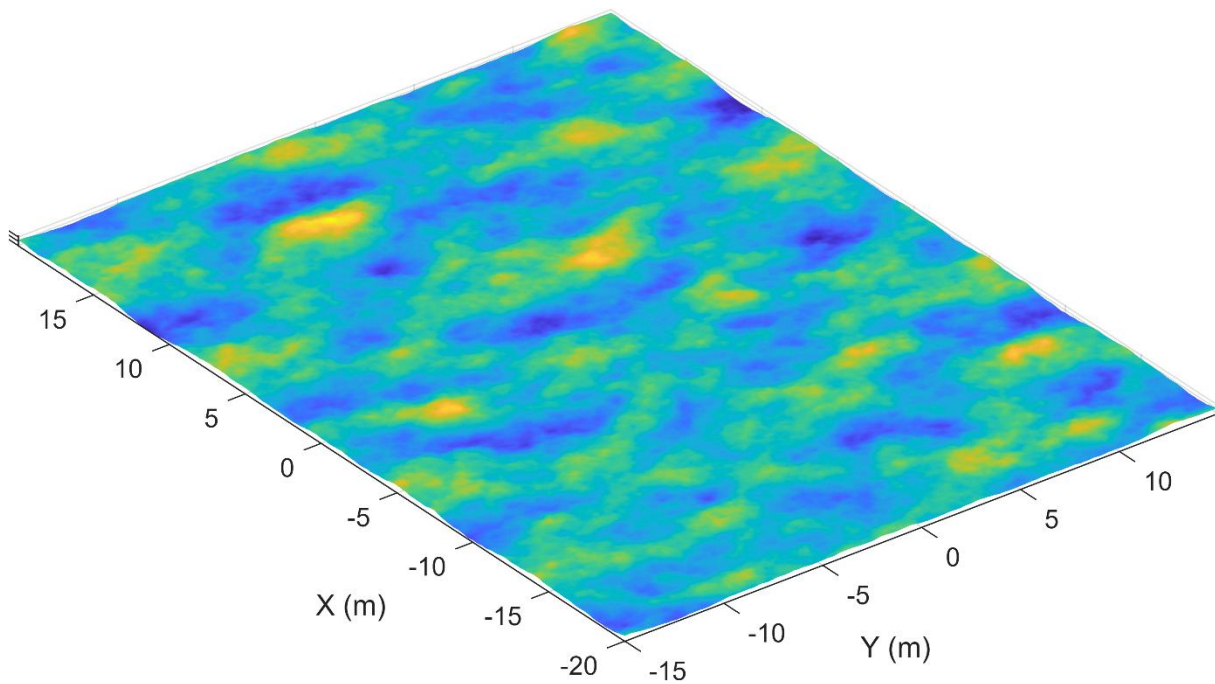


Figure 8: Tab 2 - Generated surface